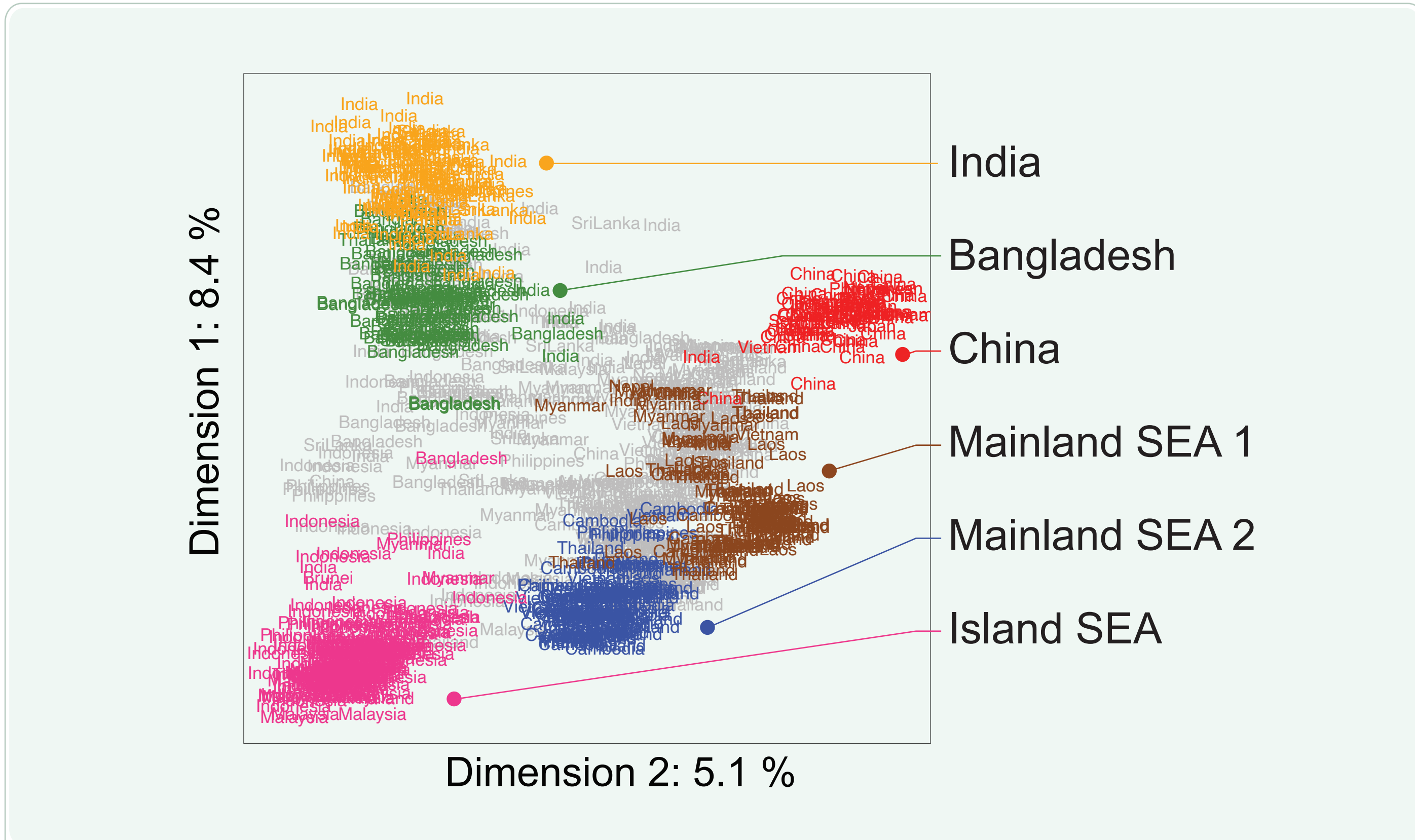


BACKGROUND

Using genome data from 957 *indica* landraces, we identified 6 geographical populations and reconstruct how *indica* rice moved from the Indian subcontinent to other parts of Asia.

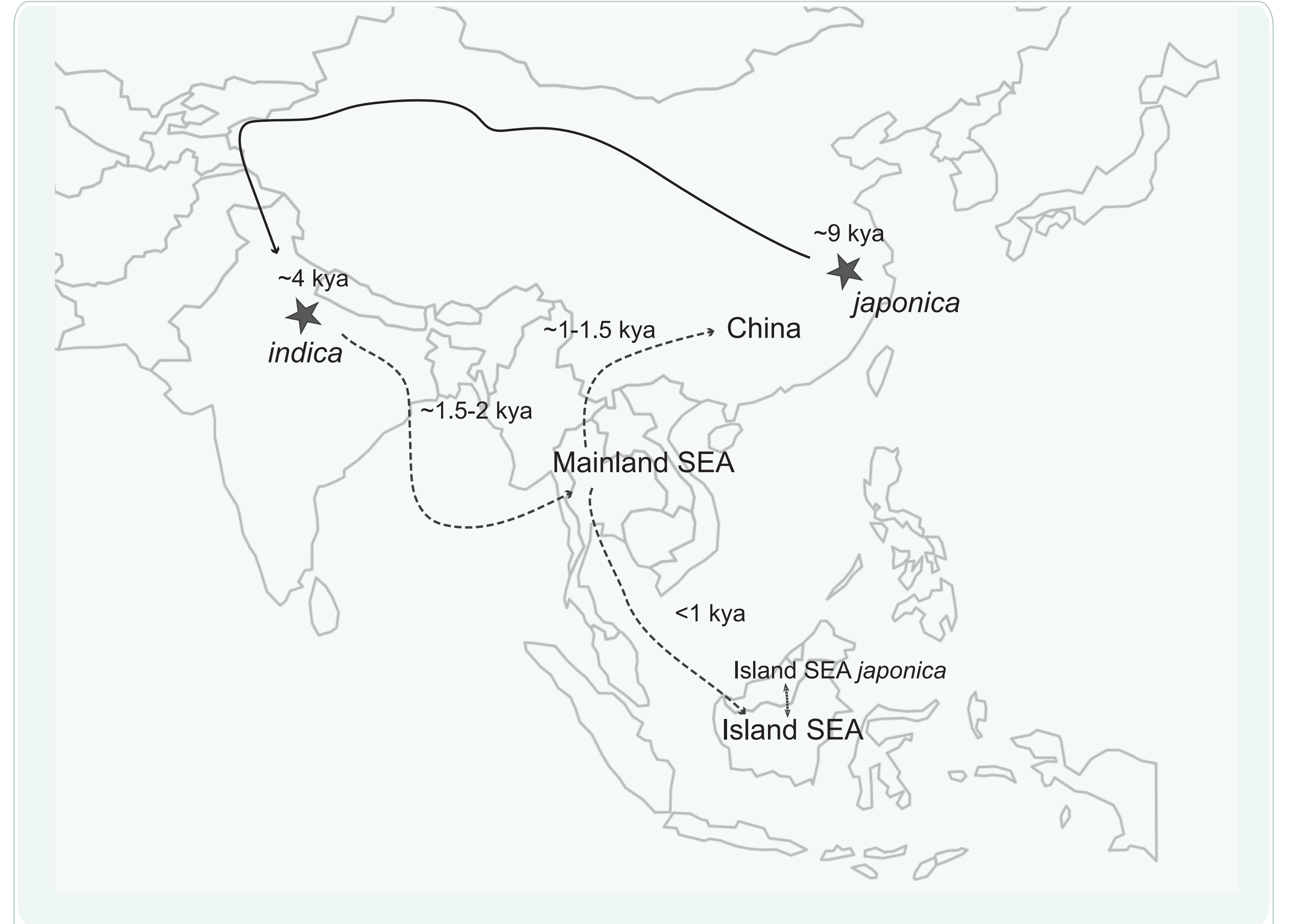


Projection of 957 *indica* landraces colored by assigned populations (k=6) onto the first two dimensions after multidimensional scaling of genomics distances.

STUDY AIMS

- ① Construct a geographically informed *indica* rice pangenome
- ② Identify genomic basis of local adaptation via pangenome

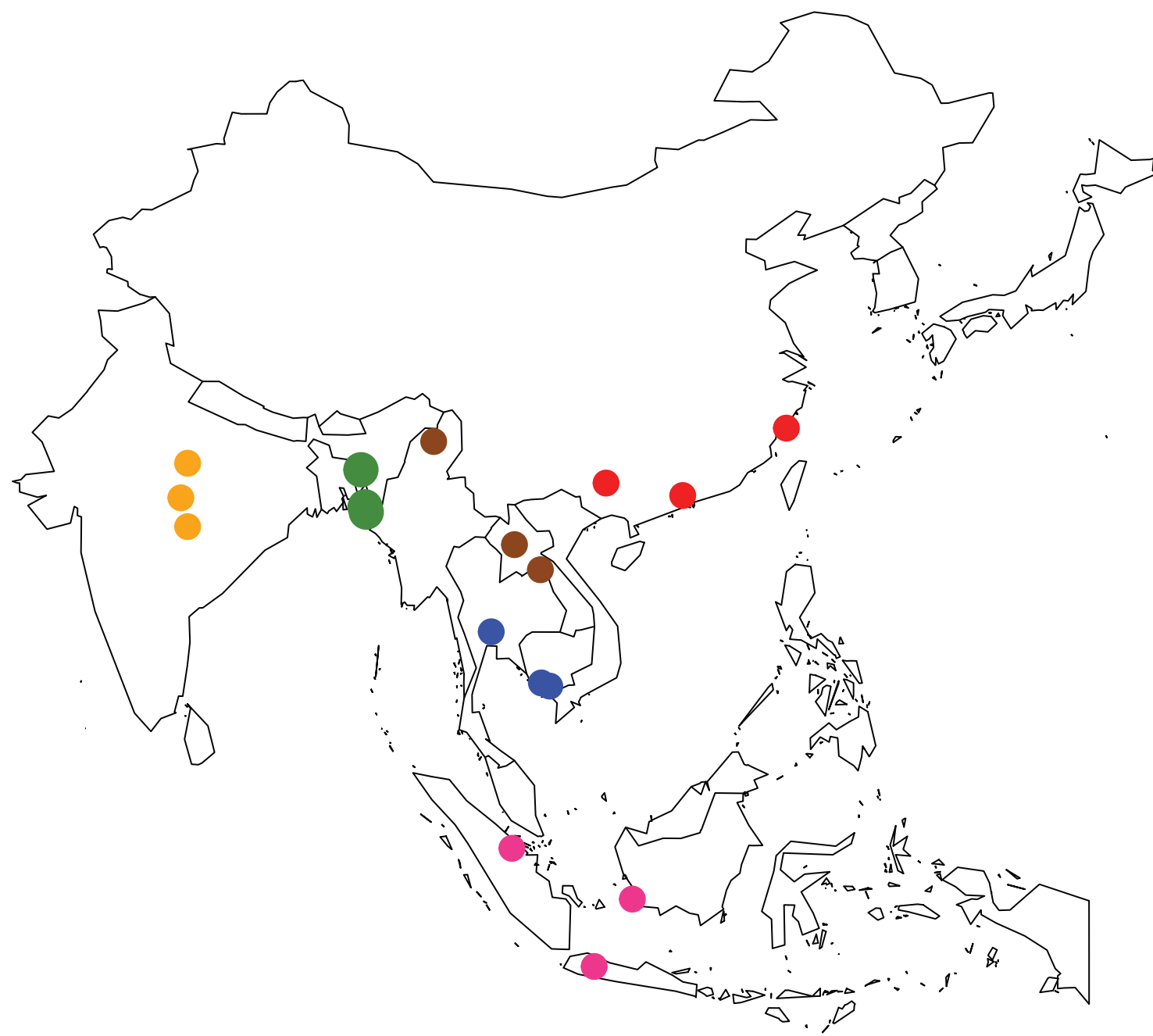
Proposed model for the spread of *indica* rice



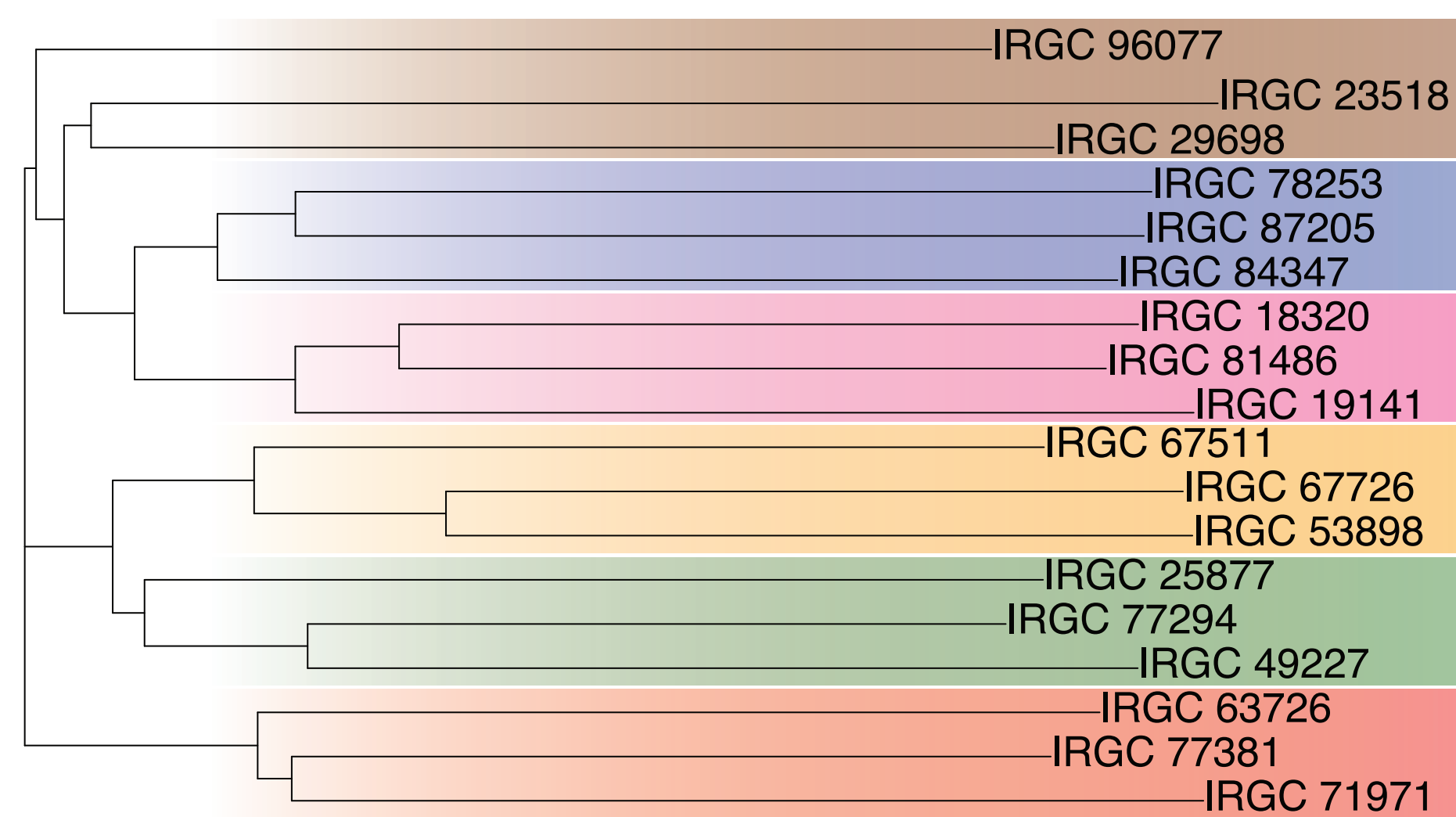
Stars indicate the domestication of *japonica* (~9,000 years BP) and *indica*, which arose with introgression (dashed line) from *japonica* (~4,500 years BP). *Indica* rice subsequently spread to Southeast Asia (SEA) and China; estimated split times are indicated in years BP. A double-sided arrow indicates gene flow between island SEA *indica* and *japonica* rice.

MATERIALS & METHODS

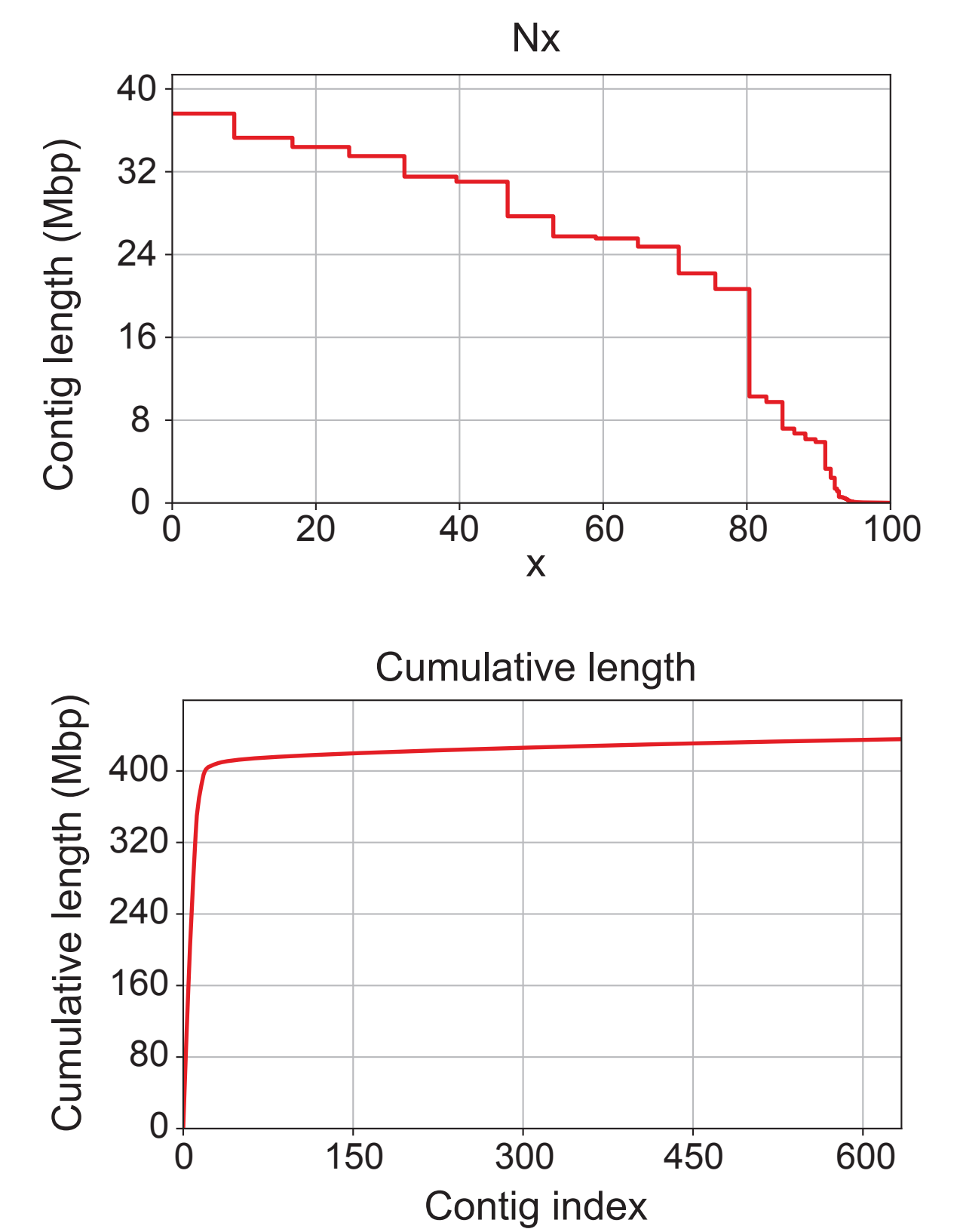
We selected 18 *indica* landraces (3 from each populations) for long-read whole genome sequencing.



Geographical origins of selected landraces colored by assigned populations.



Phylogeny of selected landraces based on genetic distance of shared biallelic SNPs.



Nx (top) and Cumulative length (bottom) plot of IRGC67511 draft assembly using default settings from hifiasm.

WORKFLOW

01 Sequencing
Long-read WGS (PacBio HiFi)
Chromosome contact (LinkPrep)
18 accessions (3 from each subpopulations)

02 Genome assembly & annotations
hifiasm for chromosome level assemblies

03 Pangenome construction
graph-based & gene based pangenome

FUTURE DIRECTIONS

Pangenome <--> Local adaptation

- ① What pangenome variations, such as presence/absence variations (PAVs) or structure variations (SVs), are associated with phenotypic traits?
- ② SVs and local adaptation
 - Duplications
 - How does selection acts on duplicated gene pairs?
 - Does local adaptation increase positive selection on duplicated genes?
 - Inversions
 - Does inversion create non-recombination evolution modules/haplotypes that retains adaptive alleles?